

DSS5201 Data Visualisation

Course Overview

David CHEW

Semester 2 AY25/26

Aims & Objectives

This course provides an introduction to the field of data visualisation, a crucial instrument in the data science industry.

- It equips students with the skills necessary to transform raw data into meaningful information by employing statistical tools and creating visual representations.
- The course is taught using relevant statistical software and covers topics including data import, cleaning, manipulation, exploration, and visualisation.
- The course emphasizes on developing skills in critically evaluating visual displays of data, ensuring a thorough understanding of data visualisation.

Syllabus

We will be covering the following:

- Basics in Python programming
- Importing data
- Data cleaning and data manipulation
- Tidy data
- Relational data
- Principles of data visualization
- Data visualization
- Advanced topics (if time allows)

Contact Hours

- We meet every Wednesday 1900-2200 hrs in LT26.
- Lectures will be recorded and made available on [Canvas > Videos/Panopto > Web Lectures](#).

Lecturer

Dr David CHEW

Office: S16-04-05

If you need to contact me,

- please use your NUS email account to write me at david.chew@nus.edu.sg.
- DO NOT use your personal email account.
- do include the course code **DSS5201** as part of the email subject.

Consultation: In-person only (email to arrange)

Assessment

Component	Weight
Class participation	10%
DataCamp assignments	20%
Group project	20%
Final exam	50%

The final examination will held on 05 May 2026, Tuesday at 1300 hrs.

Communication

It is every student's responsibility to check their NUS official email regularly. All course announcements will be sent to your NUS email.

When writing to the teaching team, please use your NUS email with the course code **DSS5201** in the subject line.

Academic Honesty

As an NUS student, you are expected to maintain and uphold the highest standards of integrity and honesty at all times.

Lack of knowledge of the NUS Code of Student Conduct is NOT a reasonable explanation for a violation.

DataCamp Assignments

As part of the class, you are required to complete 8 DataCamp assignments. Refer to the schedule below (as well as the DataCamp portal) for the due date of each assignment.

- **Activate your account to join DataCamp Classroom by 16 Jan 2026.**
- An invitation link has been posted on Canvas.¹
- Sign in with your NUS email (e.g., e0123456@u.nus.edu) so your work can be properly tracked.
- Use the name that appears on your student card as your user/display name.

1. https://www.datacamp.com/groups/shared_links/29a618a8a75a4dba5402dd5032b0dd941ab659074585efe683cb6f

DataCamp Assignments

- Assignments will be graded based on **completion by the due dates**, not by XP points. For each assignment, you will receive a binary grade – full or none.
- All assignments and deadlines are provided quite far ahead in the syllabus. Deadline extensions are generally not given.
- Check out the following [FAQs](#) about DataCamp assignments.

LEARN

APPLY

Getting Started (3/4)

Assignments

DSS5201 (AY 25/26 Sem 2)

TITLE	ASSIGNER	DUE BY	STATUS
 Introduction to Python Course		Fri, Jan 30, 2026	COMPLETED
 Intermediate Python Course		Fri, Feb 6, 2026	COMPLETED
 Data Manipulation with pandas Course		Fri, Feb 13, 2026	IN PROGRESS
 Reshaping Data with pandas Course		Fri, Mar 6, 2026	IN PROGRESS
 Joining Data with pandas Course		Fri, Mar 13, 2026	IN PROGRESS
 Introduction to Data Visualization with Matplotlib Course		Fri, Mar 20, 2026	IN PROGRESS
 Introduction to Data Visualization with Seaborn Course		Fri, Mar 27, 2026	IN PROGRESS
 Introduction to Data Visualization with Plotly In Python Course		Fri, Apr 10, 2026	IN PROGRESS

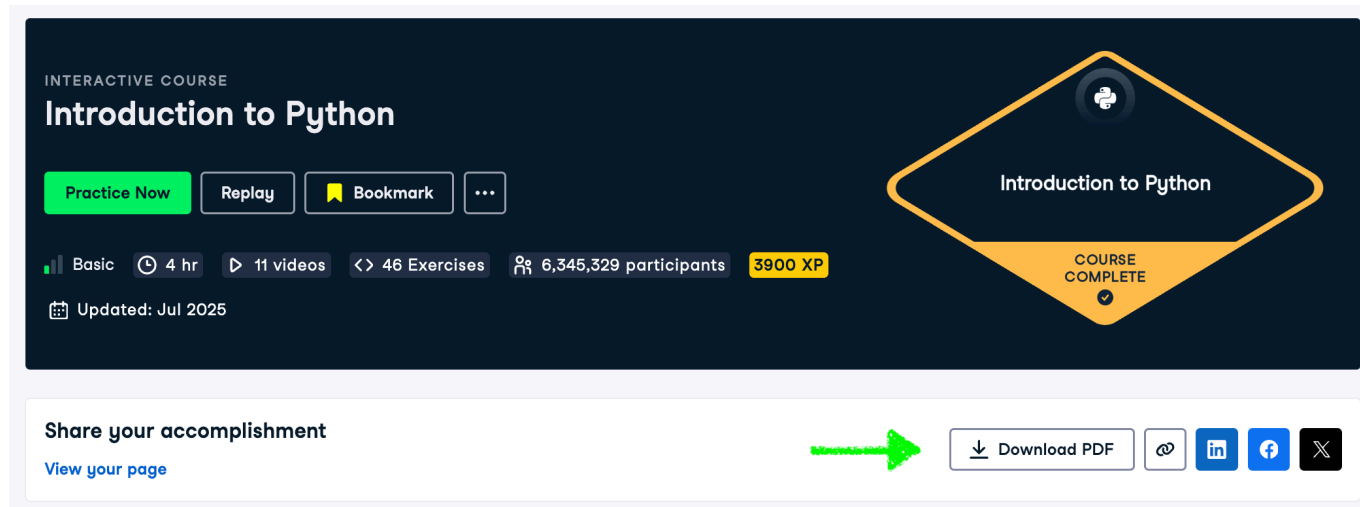


Completing an DataCamp Assignment

DataCamp assignments are due on Fridays 11:59pm (Singapore time).

Once you complete the course, you will receive a **Certificate of Accomplishment** via email.

- Make sure to receive the Certificate before the due date.
- DataCamp can be lagging at times. It's best to avoid rushing the assignment in the last minute.



Tentative Schedule

Week	Topics	DataCamp Assignments*
1	Setting up the environment	
2	Basics in Python Programming I	
3	Basics in Python Programming II	Introduction to Python
4	Importing Data with Python	Intermediate Python
5	Data Manipulation	Data Manipulation with pandas
6	NO CLASS: Chinese New Year	
	Recess Week	
7	Tidy Data; Relational Data	Reshaping Data with pandas
8	Principles of Visualization	Joining Data with pandas
9	Data Visualisation I	Introduction to Data Visualization with Matplotlib
10	Data Visualisation II	Introduction to Data Visualization with Seaborn
11	Exploring Data through Visualization	
12	Exploring Data through Visualization	Introduction to Data Visualization with Plotly in Python
13	Exploring Data through Visualization	

*All DataCamp assignments are due at 11:59 pm (Singapore time) on Fridays of the week stated.

Group project

Project topics and guideline will be posted after the Recess Week.

- You will work as a team of **4-6 members**.
- Choose among a given set of topics and data sets.

More details will be announced in time to come.

Software requirements

In this course, we will be using the following software.

1. Python
 2. VS Code
 3. Exemplify (for Final Exam)
- We will spend some time during the first lecture to help everyone set up the Python + VSCode environment.
 - You are expected to bring your laptop to class.

Seeking help

You can reach out to us (the teaching team) after classess.

- Email us using your NUS email account with **DSS5201** in the subject title.

How to ask a debugging/coding question?

- **Copy and paste your code** and the entire error message in a fixed width-font like `Courier`.
- Ideally, code that reproduces the error (`minimal working example`).
- **Do not send screenshots.**